

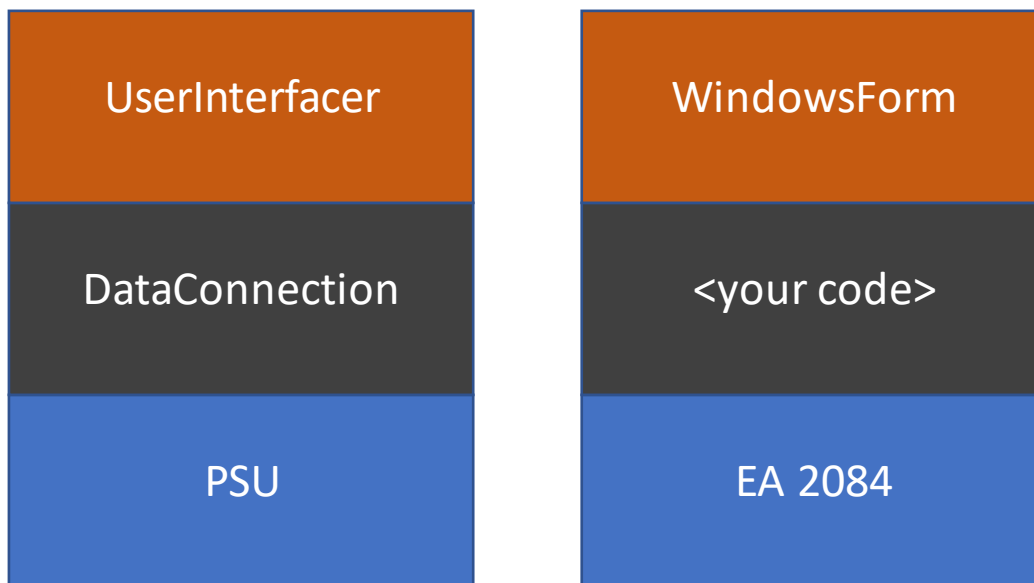
Assignment 2

In the previous assignment you made a user interface (UI) with windows forms for the power supply unit (PSU).

Before this assignment, you should label your code if you use a source control system (e.g. GIT), or make a copy of the project if you don't use a source control system.

In this assignment you shall start implementing a layered system. It shall be a three-layer system, where the 3rd layer (bottom layer) is the PSU. So you will have a front end (e.g. your windows forms application), then the code for connecting to the PSU and performing the interactions against the PSU.

Below is a general example (to the left) and a specific example for your project (to the right)



This view is a logical view, and does not tell how this layers should be implemented. One way to implement this is to let all the code that is not part of the UI belong in the gray box. That means there should be no code in the orange box (in that Visual Studio project), that knows anything related to the blue box – e.g. System.IO.Streams.

In your current solution, add a new C# project call class library. Add dependencies between the forms project and the class library.

Extension 2

So in this extension, you should use the Interface type.

Make an interface in you new class library, that hides the implementation of the PS2000 implementation. Remember that an interface have all its methods as public, and that then interface itself must be public in the library.

Once you have been able to use the interface from the GUI application, add a factory class to your library (it must be a static class) that has a static function for creating an instance of the interface type.